



Tracking Cam Prototype 0.5 PCB: Development and Testing Platform

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Revision History

Date	Version	Notes
2026-03-04	1.0	Initial design documentation

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1 Symbols, Abbreviations, and Acronyms

Symbol	Description
PCB	Printed Circuit Board
PD	Power Delivery (USB-C specification)
UART	Universal Asynchronous Receiver-Transmitter
PWM	Pulse Width Modulation
FCCM	Forced Continuous Conduction Mode
SWD	Serial Wire Debug
CRC	Cyclic Redundancy Check
MCU	Microcontroller Unit
TPS54308	Texas Instruments 3A synchronous buck converter
HUSB237	Hynetek USB PD sink controller
STM32F405	STMicroelectronics ARM Cortex-M4 microcontroller
Jetson	NVIDIA Jetson Orin Nano embedded GPU module
V_{ref}	Internal voltage reference
V_{out}	Regulated output voltage
μF	Microfarad (capacitance unit)
mA	Milliampere (current unit)
MHz	Megahertz (frequency unit)

2 Introduction

The RoCam (High Performance Vision-Guided Rocket Tracker) is a system developed for the McMaster Rocketry Team to autonomously track and record model rocket launches using a gimbal-mounted camera. This report documents the “Tracking Cam Prototype 0.5 PCB,” a development and testing platform designed to support early software development and hardware integration for the RoCam project.

Purpose: The Prototype 0.5 PCB serves as a development and testing platform that allows the team to develop and validate the control firmware, power management, and Jetson integration without requiring the full outdoor gimbal assembly. This enables parallel development tracks: software development can proceed on the lab bench while the mechanical gimbal structure is being fabricated and tested separately.

The PCB integrates three critical functions into a single compact design:

1. **Jetson Orin Nano Power Delivery:** A dedicated 17V, 2A regulator provides stable power to the NVIDIA Jetson Orin Nano, which runs the vision processing and tracking pipeline.
2. **Servo Power Delivery:** Two independent 5.5V, 2A regulators power the pan and tilt servos for benchtop testing.
3. **Gimbal Control:** An STM32F405 microcontroller manages real-time servo control and communicates with the Jetson via UART, implementing the gimbal protocol defined by the McMaster Rocketry Team.

Note on SRS Compliance: This prototype PCB is intended for *development and testing only* and does not fully adhere to the Software Requirements Specification (SRS) constraints that apply to the final deployment gimbal. For example, the SRS environmental specifications (EPE-1: 0°C to 45°C) and safety requirements apply to the deployment system, not this benchtop testing platform.

The board operates from a single USB-C Power Delivery (PD) source, enabling portable benchtop testing without requiring AC power infrastructure.

3 Design Constraints

The PCB design was constrained by system requirements, environmental conditions, safety considerations, and budget limitations derived from the project specification.

3.1 Power Budget

The total power budget is derived from the USB-C PD input capability and the loads’ requirements:

The 65W USB-C PD input provides sufficient headroom for conversion inefficiencies and transient loads.

Table 1: Power budget allocation for the RoCam PCB.

Load	Voltage	Current	Power
Jetson Orin Nano	17V	2.0A	34W
Pan servo	5.5V	2.0A	11W
Tilt servo	5.5V	2.0A	11W
STM32F405	3.3V	0.1A	0.33W
Total			56.33W
Input (PD)	20V	3.25A	65W
Headroom			8.67W (13%)

3.2 Input Power

The board accepts power from a single USB-C PD source capable of delivering 20V at 3.25A (65W). This configuration was chosen for ease of use; operators can use any USB-C PD charger or power bank capable of 65W output.

3.3 Voltage Rails

The PCB generates three regulated voltage rails:

- **17V rail:** Powers the Jetson Orin Nano through a DC barrel connector. The 17V output stays within the Jetson’s specified 7–20V input range while providing margin above the minimum to account for cable losses and transients.
- **5.5V rails:** Two independent rails for the pan and tilt servos. The 5.5V output (nominally 5.6V) provides compatibility with standard hobby servos while remaining below their typical 6V maximum rating.
- **3.3V rail:** Derived from the Jetson’s 3.3V output on the expansion header to power the STM32F405 microcontroller.

3.4 Form Factor and Mounting

The PCB dimensions are **100mm × 80mm**, matching the footprint of the NVIDIA Jetson Orin Nano Developer Kit carrier board (100mm x 79mm) (NVI, 2024). This allows the PCB to be mounted directly on top of the Jetson using standoffs, creating a compact stack suitable for benchtop development and testing without requiring the full outdoor gimbal assembly.

3.5 Safety Considerations

The onboard reset switch (SW1) controls the enable pin of the Jetson power regulator, allowing power cycling the Jetson and STM32. Note: See Section 8 for a known issue with

servo power during reset.

3.6 Budget Constraints

The total project hardware budget is limited to \$500 CAD. The PCB and components must fit within this allocation alongside the Jetson, camera, servos, and gimbal hardware. To reduce BOM cost and simplify procurement, the **same buck converter component (TPS54308) is used for all three voltage rails**—two servo rails and one Jetson rail—reducing component variety and enabling volume pricing.

4 System Block Diagram

Figure 1 presents the system-level block diagram showing power flow (black arrows) and signal flow (blue arrows).

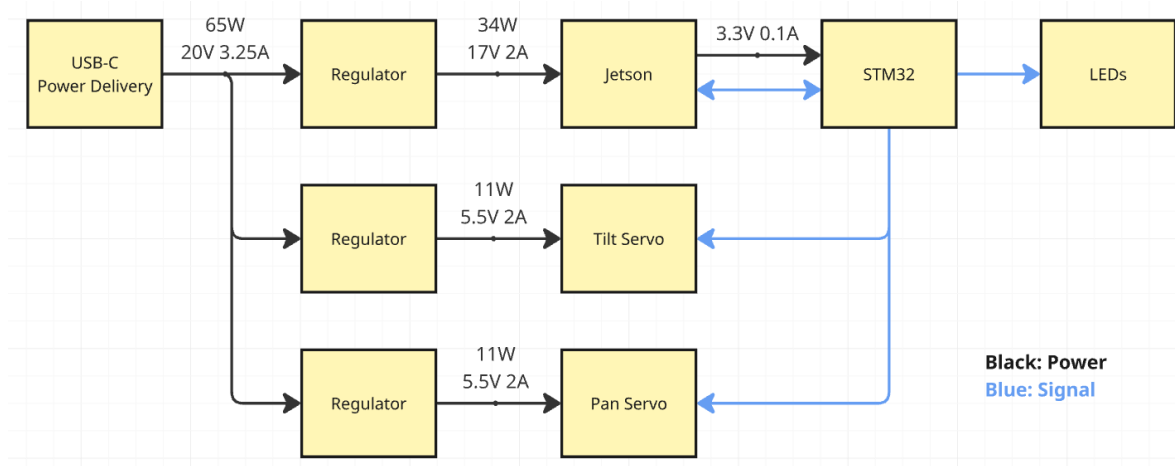


Figure 1: System block diagram for the Tracking Cam Prototype 0.5 PCB showing power delivery from USB-C PD through three independent buck converters to the Jetson (17V), tilt servo (5.5V), and pan servo (5.5V). The STM32F405 receives 3.3V from the Jetson header and communicates via UART while generating PWM signals for servo control. Black arrows indicate power flow; blue arrows indicate signal flow.

The USB-C connector accepts 20V/3.25A from a PD-compliant source. The HUSB237 PD controller negotiates the 20V contract with the source. Three independent TPS54308 buck converters step down the 20V input to the required output voltages. The STM32F405 interfaces with the Jetson via UART (PA9/PA10) for command and telemetry exchange, and drives the servos via software-generated PWM signals on PB5 (tilt) and PB6 (pan).

5 Component Selection

5.1 Buck Converter: TPS54308

The Texas Instruments TPS54308 was selected for all three DC-DC converters based on the following criteria: ([Texas Instruments, 2021](#))

- **Input voltage range:** 4.5V to 28V covers the 20V PD input with margin for transients.
- **Output current:** 3A maximum continuous output exceeds the 2A requirements for both the Jetson and servo rails.
- **Package:** SOT-23-6 enables compact layout and adequate thermal performance for the power levels involved.
- **Switching frequency:** Fixed 350kHz with Forced Continuous Conduction Mode (FCCM) provides predictable noise characteristics.
- **Internal compensation:** Eliminates external compensation components, reducing BOM complexity and board area.

The TPS54308 features an internal 0.6V voltage reference (V_{ref}) used in the feedback resistor divider to set the output voltage.

5.2 USB PD Controller: HUSB237

The Hynetek HUSB237-AA001-DN06R USB PD sink controller was selected for USB-C power negotiation: ([Hynetek Semiconductor, 2024](#))

- **Minimalist configuration:** A single external capacitor ($C23 = 20\text{nF}$) on the VSET pin selects the target 20V voltage, eliminating the need for external resistors or programming.
- **Power capability:** Supports up to 20V/5A (100W), exceeding the 65W requirement.
- **Package:** DFN-6L (2mm $\times$ 2mm) for compact layout.

The HUSB237 negotiates with the USB-C PD source to request the voltage configured by the VSET capacitor; a 20nF capacitor selects 20V.

5.3 Microcontroller: STM32F405RGIx

The STMicroelectronics STM32F405RGIx was selected for its low cost, wide availability, and compatibility with the McMaster Rocketry Team's tools and libraries. Key specifications: ([STM, 2022](#))

- **Core:** ARM Cortex-M4 with FPU at up to 168MHz
- **Memory:** 1MB Flash, 192KB SRAM
- **Peripherals:** USART, timers, GPIO, SWD debug
- **Package:** LQFP-64 for hand-solderability

5.4 Connectors

- **Jetson interface:** Samtec ESQ-106-33-L-D 6-pin header carries UART (RX/TX), 3.3V, and ground between the PCB and Jetson expansion header. The ESQ series features 0.100" (2.54mm) pitch, vertical orientation, and the Tiger Buy contact system ([Samtec, 2024](#)).
- **Servo outputs:** 3-pin 0.1" headers (signal, power, ground) for standard hobby servo connectors.
- **Jetson power:** 2-pin screw terminal for the 17V output to the Jetson's DC barrel jack.
- **USB-C:** HX TYPE-C 16P SMD 5A connector for PD input. The Hanxia HX TYPE-C 16P supports 5A current and 20V voltage rating in an SMD package ([Hanxia, 2024](#)).
- **Debug:** SWD header for programming and debugging the STM32F405.

6 Circuit Design and Calculations

This section details the circuit implementation for each functional block, including component value calculations.

6.1 USB-C Power Delivery

The USB-C power input stage uses the HUSB237 PD controller to negotiate 20V from the connected power source. Figure 2 shows the implementation.

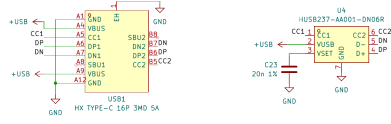


Figure 2: USB-C connector and HUSB237 PD controller schematic. The HUSB237 (U4) negotiates the PD contract; capacitor C23 (20nF) on VSET selects 20V target. USB1 is the Type-C receptacle.

The HUSB237 VSET pin uses an external capacitor to configure the target RDO (Request Data Object) voltage. The relationship between capacitance and target voltage is defined by the manufacturer; for the HUSB237, a 20nF capacitor selects 20V. This simple configuration eliminates the need for external resistors or I2C programming.

Decoupling capacitor C23 (20nF, 1% tolerance ceramic) connects between VSET and ground. A 1 μ F decoupling capacitor near the VBUS pin provides supply filtering. The CC1 and CC2 pins connect to the USB-C Configuration Channel lines for PD communication.

6.2 Servo Power Regulators

Two identical TPS54308 buck converters generate the 5.5V rails for the pan and tilt servos. Figure 3 shows one channel; the second channel is duplicated.

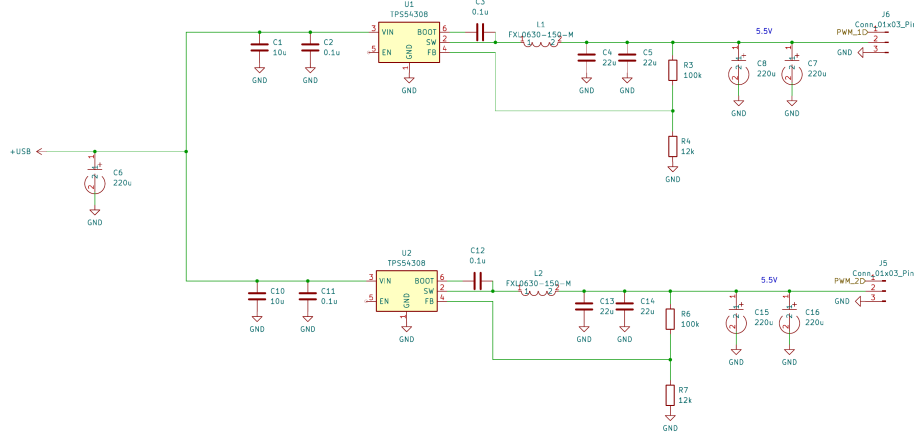


Figure 3: Servo power regulator schematic (one channel shown; two identical channels implemented). The TPS54308 (U2) steps down 20V to 5.5V with feedback divider R3/R4 and output filter capacitors C8, C7, C13, C14, C5.

6.2.1 Output Voltage Calculation

The TPS54308 regulates its output to maintain 0.6V at the FB (feedback) pin. The output voltage is set by a resistive divider:

$$V_{out} = V_{ref} \times \left(1 + \frac{R_{top}}{R_{bottom}}\right) \quad (1)$$

where $V_{ref} = 0.6V$ (internal reference).

For the servo rail ($V_{out} \approx 5.5V$):

$$R_{top} = R_3 = 100k\Omega \quad (2)$$

$$R_{bottom} = R_4 = 12k\Omega \quad (3)$$

$$V_{out} = 0.6 \times \left(1 + \frac{100k}{12k}\right) \quad (4)$$

$$= 0.6 \times (1 + 8.333) \quad (5)$$

$$= 0.6 \times 9.333 \quad (6)$$

$$= 5.6V \approx 5.5V \text{ (nominal)} \quad (7)$$

The 5.6V output provides margin above the nominal 5.5V to compensate for cable drops while remaining below the typical 6V maximum rating of hobby servos.

6.2.2 Component Selection

- **Inductor:** L1 (FXL0630-15q-M, $15\mu\text{H}$) selected for 350kHz operation at 2A load. The FXL0630-15q-M is a shielded power inductor rated for 3.5A saturation current, suitable for the TPS54308 application ([Taiyo Yuden, 2024](#)).
- **Output capacitors:** C8/C7 ($220\mu\text{F}$ electrolytic) provide bulk capacitance for transient servo loads; C13/C14 ($22\mu\text{F}$ ceramic) filter high-frequency ripple. The TPS54308 datasheet recommends ceramic capacitors at the output for stability ([Texas Instruments, 2021](#)).
- **Input capacitors:** C1 ($10\mu\text{F}$) and C2 ($0.1\mu\text{F}$) provide input filtering as recommended in the datasheet.
- **Boot capacitor:** C3 ($0.1\mu\text{F}$) between BOOT and SW pins provides gate drive voltage for the high-side MOSFET as specified in the TPS54308 datasheet ([Texas Instruments, 2021](#)).

6.3 Jetson Power Regulator

A third TPS54308 buck converter generates the 17V rail for the Jetson Orin Nano. Figure 4 shows the circuit.

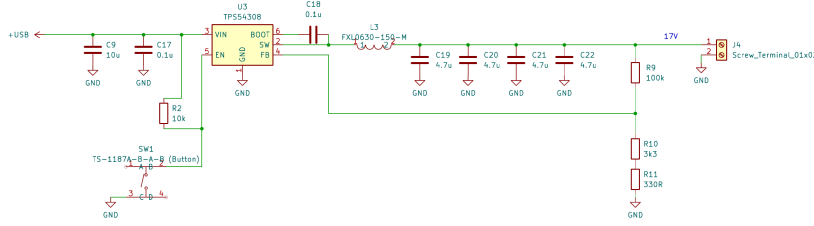


Figure 4: Jetson power regulator schematic. The TPS54308 (U3) steps down 20V to 17V with feedback divider R9/R10/R11. The EN pin is controlled by switch SW1 for power control. Output filtering uses four $4.7\mu\text{F}$ ceramic capacitors (C19-C22).

6.3.1 Output Voltage Calculation

The feedback divider uses a series combination for R_{bottom} to achieve a non-standard ratio:

$$R_{top} = R_9 = 100\text{k}\Omega \quad (8)$$

$$R_{bottom} = R_{10} + R_{11} = 3.3\text{k}\Omega + 330\Omega = 3.63\text{k}\Omega \quad (9)$$

$$V_{out} = 0.6 \times \left(1 + \frac{100\text{k}}{3.63\text{k}}\right) \quad (10)$$

$$= 0.6 \times (1 + 27.55) \quad (11)$$

$$= 0.6 \times 28.55 \quad (12)$$

$$= 17.13\text{V} \approx 17\text{V (nominal)} \quad (13)$$

The 17V output provides margin above the Jetson's minimum 7V input while remaining well below the maximum 20V rating ([NVIDIA, 2024](#)).

6.3.2 Enable Circuit

The TPS54308 EN (enable) pin controls regulator operation. Switch SW1 (TS-1187A-B-A-B tactile button) pulls the EN pin high when pressed, activating the regulator. This provides a soft-start capability for the Jetson power rail.

6.4 STM32 Microcontroller

The STM32F405RGIx microcontroller forms the core of the gimbal control system. Figure 5 shows the MCU and its supporting circuitry.

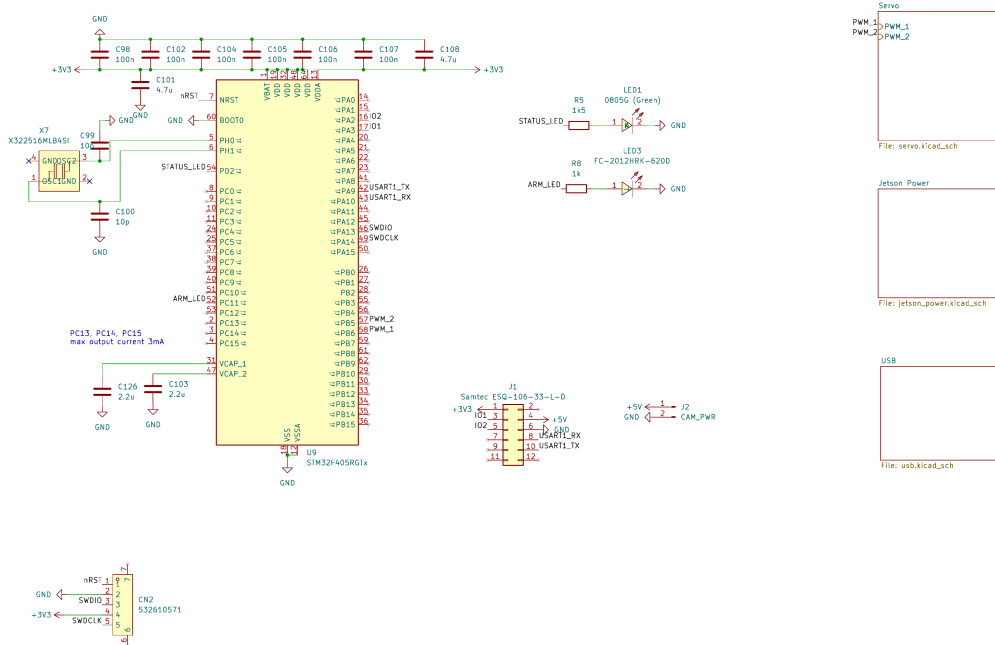


Figure 5: STM32F405 microcontroller schematic. The MCU (U9) includes decoupling capacitors (C98-C108), VCAP capacitors (C126, C103), 8MHz crystal (X7) with load capacitors, status LEDs, and the Samtec connector for Jetson interface.

6.4.1 Decoupling and Power

The STM32F405 requires extensive decoupling for stable operation at 168MHz. Per the STM32F405 datasheet, proper decoupling is essential for reliable operation (STM, 2022):

- **VDD decoupling:** Five 100nF ceramic capacitors (C98, C102, C104, C105, C106) distributed near the VDD pins filter high-frequency noise as recommended by the manufacturer.

- **Bulk decoupling:** One $4.7\mu\text{F}$ capacitor (C101) provides low-frequency energy storage.
- **VCAP capacitors:** Two $2.2\mu\text{F}$ capacitors (C126, C103) connect to the VCAP_1 and VCAP_2 pins for the internal voltage regulator. The STM32F405 reference manual specifies $2.2\mu\text{F}$ capacitors on each VCAP pin for the internal LDO stability (STM, 2022).

6.4.2 Clock Source

An 8MHz crystal (X7, X322516MLB4SI) with 10pF load capacitors (C99, C100) provides the HSE (High-Speed External) clock. The X322516MLB4SI is a 16MHz SMD crystal in a $3.2\text{mm} \times 2.5\text{mm}$ package with 9pF load capacitance specification (Yangxing Tech (YXC), 2024). The 10pF load capacitors (accounting for stray capacitance and MCU pin capacitance) ensure proper oscillation at the specified frequency. The internal PLL multiplies the 8MHz input to the system clock frequency (up to 168MHz) (STM, 2022).

6.4.3 Status Indicators and Reset

Two LEDs provide visual status indication:

- **STATUS_LED** (green, D1): Controlled via PD2; indicates system status/heartbeat.
- **ARM_LED** (red/amber, D3): Controlled via PC11; indicates armed/disarmed state.

Current-limiting resistors R5 ($1\text{k}\Omega$) and R8 ($1\text{k}\Omega$) set LED current to approximately 3mA.

The **reset button** (SW1, TS-1187A-B-A-B) is a $5.1\text{mm} \times 5.1\text{mm}$ SMD tactile switch rated for 12V DC, 50mA, with 100,000 cycle operating life (XKB Connectivity, 2024). When pressed, it pulls the regulator EN pin high to enable the Jetson power rail.

6.5 Communication Interfaces

The PCB implements two primary communication interfaces between the STM32 and external systems:

- **UART to Jetson:** USART1 (PA9/TX, PA10/RX) provides the serial link to the Jetson Orin Nano via the Samtec connector (J1). The interface supports the Gimbal Protocol as defined by McMaster Rocketry Team.
- **PWM to Servos:** Two GPIO pins (PB5 for tilt, PB6 for pan) generate PWM signals for servo control via the 3-pin servo headers.

7 PCB Layout

Figure 6 shows the PCB layout for the Tracking Cam Prototype 0.5 PCB.

7.1 Layout Considerations

- **Power traces:** High-current paths (20V input, 17V output, 5.5V outputs) use wide traces with thermal relief for component pads.
- **Ground plane:** A solid ground plane on the bottom layer provides low-impedance return paths and EMI reduction.
- **Decoupling placement:** Capacitors placed close to IC power pins minimize loop inductance per manufacturer recommendations ([Texas Instruments, 2021](#); [STM, 2022](#)).
- **Switching noise:** The TPS54308 switch node (SW) traces are kept short to reduce radiated EMI.
- **Connector placement:** USB-C, servo headers, and Jetson interface placed on board edges for cable management in the stacked configuration.
- **Mounting holes:** Four mounting holes with copper keepouts align with the Jetson mounting pattern for secure stacked assembly.

7.2 Physical Board

Figure 7 shows the manufactured PCB before assembly. The board features a black solder mask with white silkscreen labels identifying connectors and test points. The component placement follows the layout described above, with power converters in the upper left and the STM32F405 in the lower right.

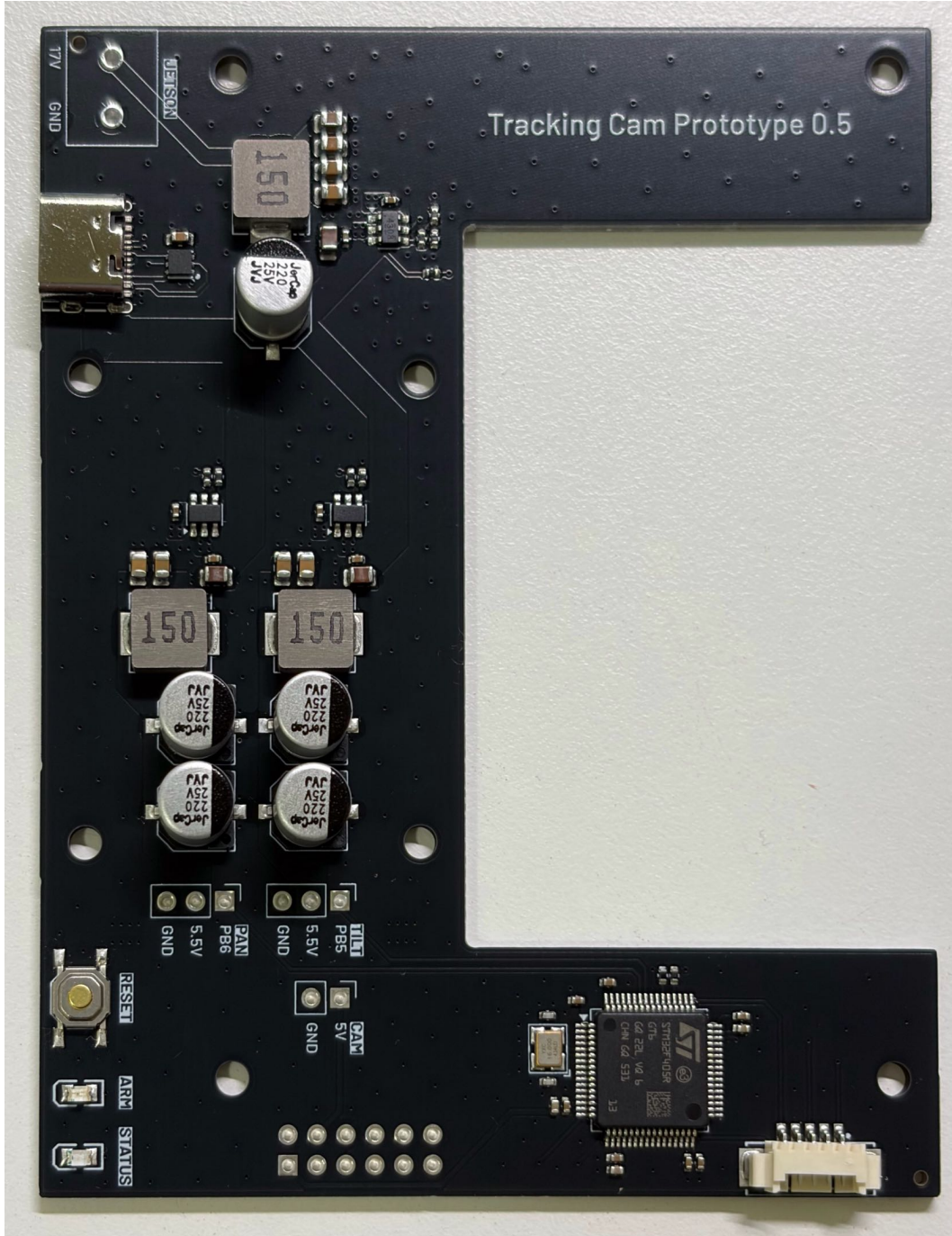


Figure 7: Manufactured Tracking Cam Prototype 0.5 PCB (bare board). The black solder mask board shows component footprints for the three TPS54308 regulators (upper left), STM32F405 (lower right), USB-C connector (left edge), and servo headers (bottom edge).

7.3 Assembled System

Figure 8 shows the fully assembled development platform. The PCB is mounted on top of the Jetson Orin Nano Developer Kit, with two hobby servos connected for benchtop testing

of the gimbal control firmware.

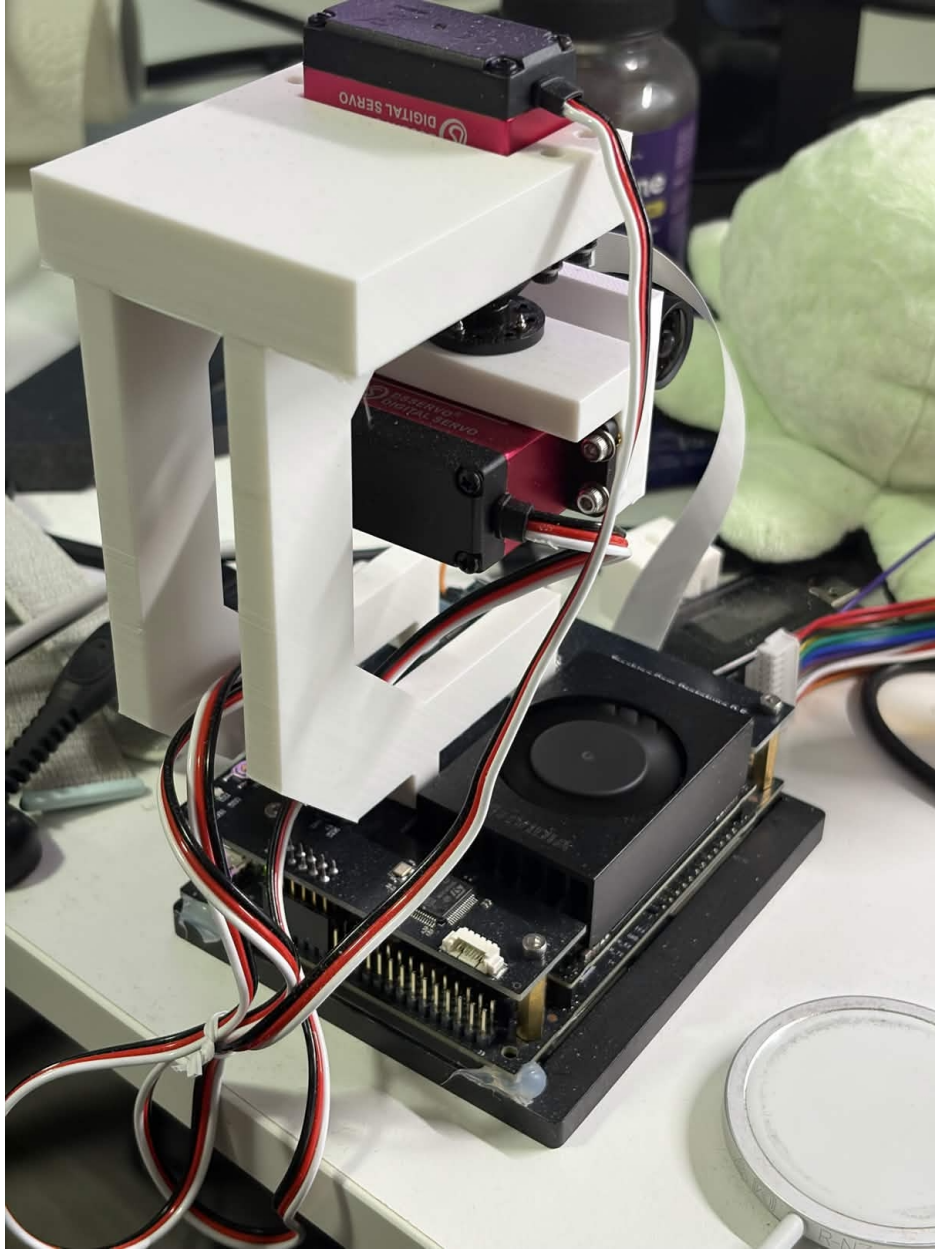


Figure 8: Assembled Tracking Cam Prototype 0.5 development platform. The PCB (top) is mounted on the Jetson Orin Nano (bottom) with standoffs. Two digital servos are connected for testing the pan-tilt control firmware on the development bench.

8 Known Issues

The following known issue has been identified in the Prototype 0.5 PCB design:

8.1 Reset Button Does Not Cut Servo Power

The reset button (SW1) only controls the enable pin of the Jetson power regulator (U3). When pressed, it cuts power to the Jetson and STM32 microcontroller, but **the servo regulators (U2) remain powered** because they are enabled by default (EN pins tied high).

This creates a potential hazard: when the STM32 is powered off but the servos remain powered, there is a small chance that electrical noise or signal transients on the PWM lines could be misinterpreted by the servos as valid commands, causing unexpected movement.

Mitigation: For benchtop testing, ensure servos are in a safe position before pressing reset. In a future revision, the servo regulator EN pins will be tied to the same reset signal as the Jetson regulator to ensure all power rails are cut simultaneously.

9 Conclusion

The Tracking Cam Prototype 0.5 PCB provides a compact development and testing platform for the RoCam project. The design delivers:

- **65W USB-C PD input** with 20V negotiation via the HUSB237 controller.
- **Three regulated rails:** 17V/2A (Jetson), 5.5V/2A (pan servo), 5.5V/2A (tilt servo).
- **STM32F405-based control** with UART interface to Jetson and PWM outputs for servo testing.
- **Stacked form factor** (100mm × 80mm) that mounts directly on the Jetson Orin Nano for compact benchtop use.

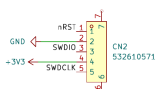
All feedback resistor values have been calculated and verified against the TPS54308 0.6V reference voltage. The use of a single buck converter component across all rails reduces BOM complexity and cost. The board is ready for assembly and benchtop firmware development for the RoCam project.

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This appendix presents the complete schematic set for the Tracking Cam Prototype 0.5 PCB.



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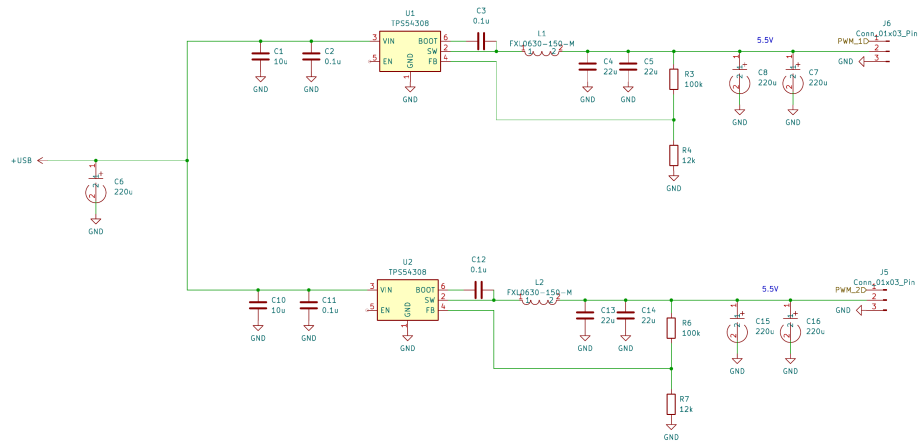


Figure 10: Servo power regulators schematic (Sheet 2 of 4).

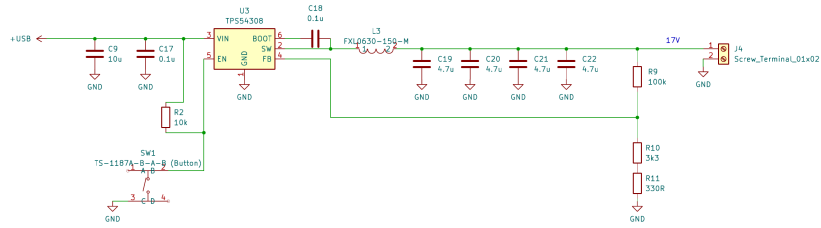


Figure 11: Jetson power regulator schematic (Sheet 3 of 4).

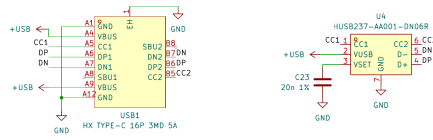


Figure 12: USB-C connector and PD controller schematic (Sheet 4 of 4).